

## CARGOMULE · PROBLEM STATEMENT

# CargoMule problem statement

Design Molecule Lab · designmolecule.com

## DOCUMENT CONTROL

|             |                                        |         |                |
|-------------|----------------------------------------|---------|----------------|
| Document ID | CGM-PRB-001                            | Version | 0.6            |
| Status      | <b>Draft</b>                           | Date    | 2026-10-02     |
| Author      | Amish Chadha                           | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept          |         |                |
| Repository  | github.com/BoujeeEnjinia1701/cargomule |         |                |
| Source      | docs/01-problem.md @ uncommitted       |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                 |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 0.6     | 2026-10-02 | Amish Chadha | Open questions O1 and O2 answered by Amish's 2026-10-02 decisions (first candidate user type; private ground only until a legal check) |
| 0.5     | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target; cost wording only, no number changed                                                     |
| 0.4     | 2026-09-26 | Amish Chadha | Stronger sources                                                                                                                       |
| 0.3     | 2026-09-25 | Amish Chadha | TRL 3. Pitch wording "most bicycles" and \$1,000 budget per CGM-DDR-001; open questions linked to the decision record                  |
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (users, context, constraints, out of scope, prior work with sources)                                                 |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                                       |

# CargoMule problem statement

Small businesses and households need to move 100 to 150 kg loads a few kilometers at a time, but a mainstream cargo e-bike lists at about \$2,400, the few electric-assist trailers on the market cost as much or more, and an ordinary bicycle cannot pull that weight up a hill. CargoMule aims to turn the bicycle a person already owns into a 150 kg load carrier for well under the price of a cargo e-bike.

## ■ The problem

Short urban and suburban trips with heavy loads (a market stall's stock, tools and materials for a trade job, a week of groceries for a large household, a delivery round) are usually made by car or van. A bicycle with a cargo trailer can do many of these trips, but the rider becomes the limit:

1. **Hills.** On level ground most people can pull about 137 kg (300 lb) on a trailer at modest speed, but a 2 % grade already triples the effort and a 4 % grade needs very low gearing ([Bikes At Work](#)). A first-order estimate for CargoMule's design load (about 188 kg of loaded trailer) on an 8 % grade is about 165 N of extra pull, or about 370 W at 8 km/h on top of the rider's own climb. That is beyond most riders.
2. **Cost of the alternatives.** A mainstream electric longtail such as the RadWagon 5 lists at \$2,399 ([Rad Power Bikes](#)). Commercial electric-assist trailers exist but are priced for businesses (see Prior work).
3. **Braking.** A heavy trailer without its own brakes pushes the bicycle when it slows. Voluntary trailer standards set low limits for unbraked trailers: 60 kg in EN 15918 and 45.4 kg (100 lb) in ASTM F1975 ([Wikipedia summary of both standards](#); [EN 15918 catalog entry](#)). A 150 kg trailer therefore needs its own brakes.

CargoMule is an electric-assist cargo trailer that hitches to most bicycles with no wiring to the bike. A load cell in the drawbar measures how hard the bicycle is pulling and the trailer's hub motor pushes in proportion, so the rider feels a fraction of the trailer's load on the flat and on hills.

## ■ Users and context

| User                                            | Need                                                              | Context                                                    |
|-------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------|
| Market trader or street vendor                  | Move 80 to 150 kg of stock and a stall between storage and market | Early starts, mixed roads, curbs, parking at the stall     |
| Tradesperson (gardener, painter, bike mechanic) | Carry tools and materials to jobs within about 10 km              | Daily use, loads vary through the day, hills               |
| Household without a car                         | Weekly shopping, moving furniture, bulk goods                     | Occasional use, stored in a shed or hallway                |
| Community group, food bank or repair café       | Pick up and deliver donations; share the trailer among members    | Several riders and bikes; must hitch quickly to any bike   |
| Local workshop or maker                         | Build and repair the trailer from common parts                    | Hand tools, a welder in some cases, e-bike parts suppliers |

## Operating environment

- **Roads:** paved streets, bike lanes and compacted gravel paths, with curbs, drop kerbs and grades up to about 8 % on short climbs (10 % occasionally).

- **Loads:** 100 to 150 kg of payload on the deck; towing bicycle and rider about 90 to 110 kg.
- **Speed:** 10 to 20 km/h cruising; assisted speed limited to 25 km/h in line with pedelec rules.
- **Climate:** –10 to 40 °C, rain, road spray and winter salt in some regions.
- **Charging:** household mains at home or at the business; charged overnight.

## ■ Constraints

- Garage-buildable prototype, with a \$1,000 USD value-engineering target for parts ( `project.yaml` , a hypothetical control target, not a limit; raised from \$900 by CGM-DDR-001, D1).
- Hitches to most bicycles with no change to the bike and no wiring to it: bikes with a quick-release or thru-axle rear wheel first, with adapters for others (see CGM-REQ-001, R7).
- Assist within e-bike limits: 250 W rated power and assist to 25 km/h at most, matching the EU pedal-assist exclusion in Regulation (EU) No 168/2013 ([EUR-Lex](#)) and within the US federal low-speed electric bicycle definition of under 750 W and 20 mph ([15 U.S.C. 2085](#)). Whether a motorized trailer falls under these rules is an open question.
- The trailer must brake itself so it never pushes the bicycle hard when slowing.
- Generic, replaceable parts (e-bike hub motor, controller, disc brakes, 20 in wheels) so a local bike shop can repair it.

## ■ Out of scope

- A complete cargo bicycle or tricycle (see FlatTrike).
- Carrying people or children.
- Handcart or walk-behind mode (kept out of scope by CGM-DDR-001, D10).
- Designing the battery cells or BMS (a bought-in 36 V LiFePO4 pack, CGM-DDR-001 D2; a SwapCell receiver is a later fleet variant).
- Road registration or type approval.

## ■ Prior work

- **Carla Cargo (Germany).** A heavy-duty trailer with an optional 250 W hub motor, 150 kg payload in its original version and an overrun brake with dual discs ([New Atlas](#)). Its current eCARLA carries up to 200 kg, uses a Bafang hub motor and a 720 Wh LiFePO4 pack, assists to 25 km/h and starts at €6,490 ([Carla Cargo](#)). The early version sensed pedaling through a crank sensor fitted to the towing bike, which ties the trailer to one bike.
- **Biomega Ein.** A single-wheel electric trailer with a 250 W hub motor that detects the bike's motion from its own wheel speed and pushes in proportion; about \$1,428 planned retail ([New Atlas](#)). It is a light-load design.
- **Roland PAXTER e.** A box trailer with two 125 W, 36 V hub motors, a 180 Wh pack and a pedal sensor, sold at about €3,300 ([Notebookcheck](#)).
- **Brake-controlling couplings.** A Bruggli patent application describes a trailer coupling that senses relative movement between two housing parts to control the trailer's brakes ([WO2022223692A1](#)), and a German utility model covers a bicycle trailer coupling for an overrun brake ([DE202010010674U1](#)). These confirm that sensing at the coupling works for braking. CargoMule uses a load cell at the coupling for propulsion as well, and a freedom-to-operate check is needed before release.

- **Unpowered heavy trailers.** Commercial cargo trailers carry 14 to 140 kg ([Wikipedia](#)), and heavy-haul builders show 137 kg is a practical unassisted limit on level ground ([Bikes At Work](#)).

The gap CargoMule addresses is an open, garage-buildable design that senses pull force at the drawbar, so it needs no sensor or wiring on the towing bike, carries 150 kg with its own brakes, and costs a fraction of a commercial electric trailer.

## ■ Open questions

- Is a motorized bicycle trailer legal on public roads and bike lanes in the first target country, and under which category? Carla Cargo's sale in Germany suggests a path in the EU; US state rules need checking. Decided by Amish, 2026-10-02 (CGM-DDR-001, O2): private ground only until a written legal check exists for the first partner's country.
- Which users first: a market trader group, a trades cooperative, or a community food bank? Decided by Amish, 2026-10-02 (CGM-DDR-001, O1): a group moving 50 to 150 kg loads on short, repeated urban trips on paved, hilly routes; the first candidate type to approach is a community food bank's local delivery or pantry runs, with market traders second. No partner is agreed.
- Is 150 kg the right payload, or do most target loads fall under 100 kg, which would allow a lighter trailer?
- How many target bikes have a quick-release or thru-axle rear wheel, and what share need a nutted-axle or hub-gear adapter?

## ■ User research and co-design

- [ ] Identify two or three target users (trader, tradesperson, community group) and record their typical loads, routes and storage space
- [ ] Ride typical routes with a loaded unpowered trailer and log grades and speeds
- [ ] Validate payload, range and cost assumptions with users
- [ ] Revise requirements (REQ) from findings before freezing the design